

One-Class SVM for PFAS–MOF Descriptor Modeling

Problem framing

- **Goal:** Learn the *region* of descriptor space occupied by known PFAS-adsorbing MOFs (positive-only data).
 - **Why OC-SVM:** No reliable negatives or scalar target; need a soft boundary around positives.
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Descriptor set

Core (chemist-prioritized):

- PLD (Å)
- LCD (Å)
- Geometric Surface Area (m²/g)
- Void fraction
- Pore volume (cm³/g)

Candidate / exploratory:

- Density (g/cm³)

Dimensionality: 5–6 features is appropriate for OC-SVM with 23 and 13 samples each for PFOA and PFOS respectively.

Modeling choices

What the OC-SVM is optimizing (intuition)

- The one-class SVM learns a *compact region* of descriptor space that contains most of the positive data.
- Unlike binary SVMs, there is no opposing class pushing the boundary from the outside.
- The solution is therefore determined by a trade-off between:
 - Minimizing the volume (or norm) of the enclosing region, and
 - Allowing a controlled fraction of points to lie outside the boundary.
- This trade-off is governed primarily by the parameter ν , which prevents the trivial solution of an infinitely large enclosing region.
- **Single multivariate OC-SVM** is standard; >2 features is fine (2D examples are for visualization only).

Kernel choice (justification)

Radial Basis Function (RBF) is preferred for one-class problems in materials descriptor spaces.

Geometric rationale:

- The OC-SVM objective is to *enclose* the region occupied by positive examples.
- **RBF kernels induce closed, flexible boundaries** in feature space that can wrap around clusters of points.
- This matches the scientific goal: learning a *compact support* of PFAS-compatible MOFs.

Kernel: RBF (default for nonlinear boundaries in materials space).

- **ν (nu):** Controls allowed outliers and acts as a regularization parameter that prevents the enclosing region from expanding indefinitely.
 - **γ (gamma):** Set via median pairwise distance heuristic or limited grid search.
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Critical preprocessing

- **Scaling is mandatory.** Use z-score or robust scaling.
 - Rationale: OC BF kernel relies on Euclidean distances; unscaled features (e.g., GSA) will dominate.
 - Correlated descriptors (e.g., PLD, LCD, pore volume) are acceptable: the OC-SVM operates on the induced distance geometry rather than assuming feature independence.
 - PCA is used strictly for visualization and interpretation, not as a required preprocessing step for training.
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Assessing descriptor influence and sensitivity

There is no native coefficient-based importance. Use **sensitivity analyses**:

1) Leave-One-Feature-Out (LOFO)

- Retrain after removing each feature.
- Compare:
 - Fraction of training points flagged as outliers
 - Changes in anomaly score distribution
 - Boundary stability

2) Perturbation sensitivity

- Perturb one feature (± 5 –10%) and observe score changes.

- Interpretable for domain experts.
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What not to do

- Do not report p-values or regression-style coefficients.
 - Do not assume monotonic effects.
 - Do not over-tune to force perceived importance.
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Visualizing the learned region

Note: Different choices of slice (e.g., PCA plane vs. feature-pair slices, or different fixed reference values) may lead to different apparent boundaries; such variations reflect the high-dimensional nature of the learned region rather than model instability.

- Any 2D decision boundary plot represents a **conditional slice** of the learned high-dimensional region.
 - PCA-based plots show the intersection of the decision function with the PCA subspace.
 - Original-feature plots vary two selected descriptors while fixing all remaining features at their mean values.
 - These visualizations are illustrative and should not be interpreted as complete projections of the full decision boundary.
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Recommended workflow

1. Train RBF OC-SVM on core features (scaled).
 2. Validate stability (v sensitivity).
 3. Add density; compare boundary and scores.
 4. Perform LOFO and perturbation analyses.
 5. Report *boundary stability, sensitivity and interpretable contour based visualisations* (PCA or feature-pair slices), not coefficients.
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Outputs to report

- Anomaly scores (continuous)
- In/out decision at chosen threshold
- Sensitivity results supporting descriptor relevance
- Visualisations.